ATTENDANCE MANAGEMENT DASHBOARD

ALGORITHM AND DATASET

This document describes the proposed algorithm, processing logic, system flow, and dataset structure for the Attendance Management Dashboard.

1. ALGORITHM

The Attendance Management Dashboard follows a structured data-processing algorithm that converts raw attendance records into attendance percentages, key performance indicators, visual summaries, and dashboard insights.

1.1 Step-by-Step Procedure

1. Collect attendance records containing student, subject, date, class/section, and attendance-status information.
2. Import the attendance data from a structured Excel or CSV file.
3. Inspect the dataset and identify missing, duplicate, or inconsistent records.
4. Clean the data by correcting inconsistent values and handling invalid or missing records.
5. Standardize fields such as Student ID, class/section, subject, date, and attendance status.
6. Transform the cleaned data into an analysis-ready structure.
7. Calculate present count, absent count, total classes, and attendance percentage.
8. Generate KPIs such as total students, average attendance, present count, absent count, and low-attendance count.
9. Summarize attendance by student, subject, class/section, and time period.
10. Create appropriate charts, tables, and KPI cards.
11. Apply interactive filters for student, subject, class/section, and time period.
12. Present the processed information through the Attendance Management Dashboard.
13. Validate dashboard calculations against the prepared attendance data.

1.2 Logic / Method Used

The project uses descriptive data analysis and attendance-percentage calculation rather than a machine-learning training model. For each student or selected group, the system determines the number of classes attended and the total number of classes recorded.

Attendance Percentage = (Number of Classes Attended / Total Number of Classes) × 100

The calculated values are aggregated and presented through KPI cards, charts, tables, and interactive filters. A defined attendance threshold can be used to identify students whose attendance requires attention.

1.3 Model / Technique Applied

The proposed project applies data cleaning, transformation, aggregation, KPI calculation, descriptive analysis, and data visualization techniques. Microsoft Excel or CSV can be used for source data and

Microsoft Power BI can be used as the Business Intelligence visualization platform. No machine-learning model is required for the basic attendance-monitoring dashboard.

1.4 Process Flow of the System

Attendance Data Collection → Data Import → Data Cleaning → Data Transformation → Attendance Calculation → KPI Generation → Data Aggregation → Visualization → Interactive Dashboard → Attendance Insights

2. DATASET

The proposed dataset contains structured student attendance records. Because the dashboard has not yet been implemented, an exact record count is not claimed in this document.

2.1 Source of Data

The attendance dataset can be prepared from academic attendance records, classroom attendance sheets, or a demonstration attendance dataset created for the project. The data can be maintained in Microsoft Excel and exported or saved as CSV when required.

2.2 Number of Records / Samples

Number of records: To be finalized during implementation based on the attendance dataset used. No fixed record count is claimed before the actual dataset is prepared.

2.3 Features / Attributes

Column / Feature	Description	Data Type
Student_ID	Unique identifier of the student	Text / Integer
Student_Name	Name of the student	Text
Class_Section	Class or section	Text
Subject	Subject name or code	Text
Attendance_Date	Date of attendance	Date
Attendance_Status	Present or Absent	Categorical
Faculty / Instructor	Faculty associated with the subject, if available	Text

2.4 Training and Testing Data Split

A training and testing split is **not applicable** to the basic Attendance Management Dashboard because it is a descriptive analytics and visualization project rather than a predictive machine-learning model. The prepared attendance dataset can be used for calculating measures and generating dashboard visualizations. If a future version introduces a predictive ML component, the dataset can then be divided into training and testing subsets.

2.5 Data Format / Type

The attendance data will be stored in a structured tabular format. Microsoft Excel (.xlsx) and CSV (.csv) are suitable source formats. Date fields will use date format, student and subject information will use text or identifier fields, attendance status will use categorical values such as Present and Absent, and calculated counts and percentages will be numeric measures.

3. EXPECTED OUTPUT

The expected output is an interactive Attendance Management Dashboard containing KPI cards, charts, tables, and filters. It is expected to help users understand overall attendance, compare subjects or classes, examine student-level performance, observe attendance trends, and identify students with low attendance.

4. SUMMARY

The proposed algorithm provides a clear path from raw attendance records to useful dashboard insights. The dataset is organized around student, class, subject, date, and attendance-status attributes. Since the project focuses on monitoring and visualization rather than machine-learning prediction, a training/testing split is not required for the basic system.